Manendra Pal Singh

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LinkedIn | GitHub | Portfolio

Summary

Senior Software Developer with 4+ years of experience building and operating cloud-native systems across wealth technology, digital payments, mobility, lending, and decentralized infrastructure. Experienced in Go, TypeScript, Kubernetes, Terraform, GitOps, AI-assisted engineering tools, and multi-cloud environments spanning AWS, Microsoft Azure, and GCP. Proven ability to develop high-scale backend services, automate production operations, improve platform resilience, and build AI-powered tools for release management and proactive incident response.

Technical Skills

Languages: Go, TypeScript, JavaScript, Java

Backend & Architecture: REST APIs, Microservices, Event-Driven Architecture, Distributed Systems, System Design

AI Engineering & Automation: Azure AI Foundry, Microsoft Foundry, Cody, AI Agents, LLM-Powered Internal Tools, AI-Assisted Incident Triage, Runbook Automation

Cloud & Platform: AWS, Microsoft Azure, GCP, Kubernetes, Docker, Helm, Terraform, Flux, GitOps, CI/CD

Data & Messaging: PostgreSQL, PostGIS, RisingWave, Materialized Views, MongoDB, Elasticsearch, Kafka, RabbitMQ, Redis

Observability & Reliability: Datadog Synthetic Monitoring, Prometheus, Grafana, OpenTelemetry, Distributed Tracing, Structured Logging, Alerting, Incident Response, Disaster Recovery

Security: OpenFGA, Fine-Grained Authorization, OAuth 2.0, OIDC, JWT, RBAC

Frontend: React.js, Redux, Material UI, Tailwind CSS, HTML5, CSS3

Testing & Tools: Go Testing, Jest, Unit and Integration Testing, A/B Testing, SonarQube, Git, GitHub, Bitbucket, Postman, Swagger/OpenAPI

Professional Experience

Alpheya — Hybrid
Senior Software Developer — April 2026 – Present

- Operate a multi-tenant SaaS platform comprising 10 services across five Azure environments, maintaining 99.99% availability through deployments across three availability zones and a dedicated disaster-recovery environment.
- Achieved recovery within 15 minutes for zonal failures and within four hours for disaster-recovery scenarios by strengthening platform resilience, failover procedures, and recovery processes.
- Designed and built an AI-assisted release-management tool from scratch using Azure AI Foundry, providing a user interface that generates and manages Helm and Terraform configurations and supports biweekly production releases.
- Designed and built a Cody-powered AI SRE tool that consumes Datadog Synthetic Monitoring results, service logs, and metrics to detect and triage service degradation before it is reported by users.
- Shifted incident handling from reactive to proactive by enabling the AI SRE tool to act as L1 support, diagnose outages, select relevant operational runbooks, and initiate approved remediation actions.

CoffeeBeans Consulting LLP — Hyderabad
SDE II - Consultant at NPCI — September 2025 – April 2026

- Enabled interoperable EV charging across 18 leading Indian charge-point operators by building a decentralized charging network on the Beckn Protocol.
- Protected the platform from telemetry spikes and downstream overload by developing Go services that apply backpressure while processing data from a decentralized, distributed network.
- Enabled reliable asynchronous communication across charging participants by implementing event-driven API workflows using Kafka and RabbitMQ.
- Onboarded legacy charging platforms by developing Beckn-compliant adapters and multi-tenant services used by BHIM EV and BHIM AI.
- Enabled location-based charger discovery by building geospatial APIs using PostgreSQL and PostGIS.
- Secured network participation by designing a multi-tenant OAuth 2.0 and OIDC authentication service supporting token issuance, validation, refresh, and tenant isolation.
- Improved production troubleshooting by implementing distributed tracing, structured logging, metrics, and alerting across multi-party transaction flows.

CoffeeBeans Consulting LLP — Bengaluru
Software Engineer - Consultant at Gojek — December 2024 – September 2025

- Reduced One-Tap Ride Booking response latency to 10 milliseconds by optimizing backend processing and implementing resilient session-resume and return-context handling.
- Improved scalability and reduced infrastructure costs for services handling 10,000–20,000 requests per minute by migrating three production services from virtual machines to Kubernetes.
- Enabled secure payment processing at approximately 2,000 transactions per minute by integrating transport services with payment gateways and implementing idempotent Mini App transaction workflows.
- Supported production-critical ride-booking systems by collaborating with product, frontend, and QA teams to deliver and operate high-volume features.

Sigma Infosolutions Ltd. Jodhpur
Software Developer August 2022 – December 2024

- Scaled a cloud-based Loan Origination System across three banking clients, including a primary deployment serving approximately 800 branches and 3,000 branch-portal users.
- Reduced processing time for a banking workflow from four hours to 100 minutes by redesigning its processing and orchestration logic.
- Reduced a separate workflow's execution time to approximately 10 minutes through backend and workflow optimizations.
- Improved maintainability across more than 100 APIs by designing five Go microservices for document processing, borrower lifecycle management, and workflow orchestration.
- Accelerated frontend delivery across six to seven projects by creating a reusable React and Material UI component library.
- Reduced new-client implementation time to approximately one month by standardizing shared components, APIs, validation rules, and configurable workflows.
- Secured administrative and lending workflows by implementing granular Role-Based Access Control for administrators, loan officers, and operations teams.
- Supported monthly production releases by maintaining Dockerized services, automated testing, and CI/CD pipelines.

Open-Source Contributions

- **beckn-onix**: Contributed seven public pull requests improving network observability, Redis TLS support, Go runtime performance, and test reliability.
- **ubc-ev-sandbox**: Contributed 12 public pull requests covering EV sandbox development, production configuration, schema validation, observability, Redis TLS support, and release upgrades.
- **Kestra**: Contributed a production fix to the open-source orchestration platform.

Education

M.B.M. Engineering College Jodhpur
Bachelor of Engineering in Electronics and Electrical Engineering | CGPA: 8.2/10 July 2022